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Intake 40 - Individual Research Project (IT42999)

Design and Evaluation of an Agentic AI-Driven Decision Support System for
Explainable Early Disease Risk Detection in Companion Animals

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Chapter 1. Introduction

1.1 Background of the Study

Companion animals, particularly dogs and cats, are an important part of many households, yet they cannot verbally communicate pain or illness. Early health problems often appear through subtle changes in appetite, water consumption, activity, urination, or other behaviours that pet owners may overlook or find difficult to interpret. As a result, veterinary care may be delayed until symptoms become more severe, potentially affecting treatment outcomes and animal welfare.

Although existing digital pet-care applications provide features such as health records, vaccination reminders, and general health information, many offer limited support for analysing combinations of symptoms and guiding owners towards appropriate action. Advances in Machine Learning and Agentic Artificial Intelligence provide an opportunity to address this limitation by combining risk prediction with intelligent reasoning and personalised recommendations. Therefore, this research proposes an Agentic AI-driven decision support system that analyses owner-observed symptoms and behavioural patterns, provides explainable early disease risk information, and guides pet owners towards appropriate care, including veterinary consultation and preventive health management.

1.2 Problem Statement

Companion animals often suffer from life-threatening diseases that show limited or non-specific symptoms in their early stages. Pet owners lack accessible tools to identify early disease risk based on observable behavioral and physical changes, resulting in delayed veterinary consultation and advanced disease progression.

Three deficiencies compound this problem:

1. **Interpretive deficiency** - Subtle changes in appetite, hydration, activity, and elimination are individually unremarkable and only meaningful in combination. Owners lack the clinical framework to combine them.
2. **Technological deficiency** - Existing digital pet health tools offer static information or manual checklists, lacking autonomous reasoning, real-time analysis, and actionable decision support.
3. **Continuity deficiency** - Current systems do not integrate preventive care functions such as vaccination tracking, nor do they connect risk awareness to real-time access to veterinary services, limiting their effectiveness in promoting timely intervention.

The absence of structured, intelligent risk assessment mechanisms reduces owner confidence and weakens the adoption of preventive practices.

1.3 Research Aim

To develop an Agentic AI-driven decision support system for early disease risk detection in companion animals.

1.4 Research Objectives

1. To identify early disease risk in companion animals using owner-observed symptoms and behavioral patterns.
2. To develop and evaluate machine learning-based models for disease risk prediction.
3. To implement an Agentic AI framework that autonomously reasons over risk levels and provides action guidance.
4. To integrate location-based veterinary clinic and surgeon availability with real-time time-slot information.
5. To design a vaccination management and reminder mechanism to support timely preventive care for companion animals.

1.5 Research Questions

RQ1. To what extent can owner-observable symptoms and behavioural indicators, collected through a structured mobile interface, support reliable early disease risk stratification in dogs and cats?

RQ2. Which machine learning approaches yield acceptable predictive performance on owner-reported symptom data, and what are their limitations when trained on imbalanced or partially synthetic datasets?

RQ3. How can an agentic AI layer improve upon a single predictive model by autonomously reasoning over risk, generating explanations, and triggering preventive and referral actions?

RQ4. How can predictions be made *explainable* to a non-expert owner, such that the system builds appropriate trust without encouraging self-diagnosis?

RQ5. How can a decision support system enforce appropriate privacy and access control over sensitive animal health records shared between owners and veterinary professionals?

Table 1 : Mapping of Research Objectives to Research Questions

Objective	Addressed by	Evaluation Method
O1	RQ1	Model performance on held-out symptom data; expert review sample
O2	RQ2	Accuracy, macro-F1, ROC-AUC, PR-AUC, calibration analysis
O3	RQ3	Agent trace inspection; degraded-mode behaviour; monitoring agent output
O4	RQ3	Functional testing of discovery, availability, and booking flows
O5	RQ3	Preventive-care agent behaviour; vaccination escalation logic

(cross-cutting)	RQ4	Explainability artefacts: ontology links, feature attribution
(cross-cutting)	RQ5	Access-control test matrix; role-scoped API verification

1.6 Scope of the Study

In scope

- Species limited to dogs and cats.
- Input restricted to owner-observable symptoms, behaviors, and pet demographics no laboratory values, imaging, or wearable sensor streams.
- Risk stratification into low / medium / high with an associated confidence score, plus ranked candidate conditions.
- An agentic layer producing explanation, urgency assessment, preventive-care analysis, and veterinary referral.
- Vaccination record management and risk-aware reminders.
- Location-based veterinary discovery, surgeon availability, and appointment booking.
- A trilingual (English / Sinhala / Tamil), mobile-first, installable interface.

Out of scope

- Clinical diagnosis, prescription, or treatment planning.
- Species other than dogs and cats.
- Image-based diagnosis and IoT/wearable sensor integration.
- Integration with veterinary practice management systems or national animal health registries.
- Commercial deployment, payment processing, and telemedicine consultation.

1.7 Significance of the Study

- **Animal welfare** - Enhances early disease risk detection, supporting timely veterinary intervention and reducing the severity and cost of treatment through earlier awareness.
- **Owner empowerment** - Converts diffuse worry into a structured, explainable assessment with a clear recommended action, strengthening responsible pet ownership.
- **Methodological contribution** - Demonstrates the practical application of agentic AI to veterinary decision support, moving beyond single-model prediction toward multi-agent reasoning with autonomous action.
- **Explainability contribution** - Combines a curated knowledge ontology with model feature attribution, so that predictions are accompanied by reasons rather than presented as opaque outputs.

- **Local relevance** - A trilingual, offline-tolerant, low-cost digital health tool suited to Sri Lankan socio-economic and linguistic contexts, strengthening national capacity in applied AI and digital health innovation.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews the current state of research at the intersection of machine learning, agentic artificial intelligence, and companion animal health. The review establishes what has been demonstrated, identifies the boundary of current capability, and locates the gap this project addresses. The reviewed literature spans quantitative health scoring, clinical visit classification, behavioral sensor fusion, conversational AI for pet owners, and broader surveys of AI in veterinary medicine.

2.2 Review of Related Studies

Quantitative health scoring - Kim and Kim [1] developed an AI-based health score system using sensor-collected behavioral data to quantitatively assess a dog's health status. The resulting score correlated highly with veterinary assessments (approximately 87.5% agreement), demonstrating a data-driven and owner-accessible health monitoring model. The present project extends this by integrating agentic reasoning to interpret risk patterns and trigger personalized alerts, rather than producing a score alone.

Clinical visit classification - Szlosek et al. [2] validated a machine learning model distinguishing preventive wellness visits from other veterinary visits using clinical records, achieving specificity of 0.94 and sensitivity of 0.86. This indicates real potential for subclinical disease identification during routine assessment. The present system adapts the concept to owner-reported rather than clinic-recorded data, moving detection earlier in the care pathway.

Surveys of ML in animal health - Das et al. [3] review machine learning techniques across animal health, highlighting persistent gaps in early detection and the need for real-time, accessible models. Bouchemla et al. [4] systematically review AI feasibility in veterinary medicine, identifying diagnostic and decision-support roles across haematology, imaging, and algorithmic support. Akinsulie et al. [5] survey potential applications across veterinary clinical practice and biomedical research. Collectively these establish decision support as distinct from autonomous diagnosis as the credible near-term application of AI in this domain.

Conversational AI - Jokar et al. [6] examine the opportunities and challenges of AI chatbots assisting pet owners with health information and decision-making. While not constituting a full decision support system, this work demonstrates that conversational interfaces can provide early guidance from described symptoms. The present project incorporates this capability but deliberately constrains it: the assistant routes users toward the structured assessment rather than predicting from free text, preserving a clean, evaluable prediction pipeline.

Behavioral data and sensor fusion - Aguilar-Lazcano et al. [7] provide a scoping review of machine learning-based sensor data fusion for animal monitoring, showing how behavioral data can be processed into health insight. This underpins the behavioral component of the present system's feature set, though the present work substitutes owner observation for instrumented sensing, a deliberate accessibility trade-off.

Predictive modelling of specific conditions - Imada et al. [8] compare tree-based algorithms for predicting future diagnostic test results, noting that such models function as predictive algorithms rather than standalone diagnostics, precisely the positioning adopted here. Rath and Sumathi [9] review ML classification of pet diseases including diabetes, arthritis, and kidney disease, supporting the feasibility of multi-class risk estimation. Kulkarni et al. [10] propose an ML framework for animal disease prediction using environmental, behavioral, genetic, and historical data.

Biomarkers and the early-signal problem - Eman et al. [11] survey molecular biomarker discovery for animal disease, emphasizing the central difficulty that clinical signs appear late. Although biomarkers are laboratory-centric and inaccessible to owners, this work reinforces the value of symptom and behavioral data as the only early signal available outside a clinical setting.

Integrated assistance systems - Aishwarya [12], proposed PAWPAL, combining CNN-based image disease detection with NLP symptom analysis to deliver early guidance, achieving high accuracy on common skin conditions. Jin et al. [13] examine automated symptom extraction from clinical records for disease prediction, demonstrating reduced clinician workload. Shinde et al. [14] review AI in animal disease surveillance, prediction, and diagnosis, noting challenges of data quality and system reliability. Pereira et al. [15] overview AI in veterinary imaging.

2.3 Theoretical Background

Design Science Research (DSR) - This project constructs and evaluates an artefact, a working system, as its primary research contribution. DSR provides the appropriate paradigm, framing the work as iterative build-and-evaluate cycles that generate design knowledge alongside a functioning system.

Supervised classification - Disease risk stratification is framed as a supervised learning problem. Text-derived features from structured symptom input are mapped to condition classes and to a binary danger indicator. Linear models are selected over deep architectures for interpretability, small artefact size, and CPU-only inference (NFR4).

Class imbalance and probability calibration - Where a training set is heavily skewed toward one class, an uncorrected classifier produces systematically inflated probabilities. Base-rate correction shifting a probability from its training prior to a deployment prior preserve ranking while decompressing absolute values. This is central to the honest use of Model B.

Knowledge representation via ontology - A property graph of diseases, symptoms, vaccines, and weighted relationships supplies domain knowledge the statistical models do not contain, and provides a substrate for explanation independent of model internals.

Agentic AI - Distinct from a single model invocation, an agentic system decomposes a goal across specialized agents that reason, invoke tools, and act. Orchestration here is graph-based, with deterministic edges and conditional routing, rather than an open-ended autonomous loop a choice that trades flexibility for reproducibility, which matters for academic evaluation.

Explainable AI (XAI) - Two complementary mechanisms are employed: knowledge-based explanation (ontology links between reported symptoms and candidate conditions) and *model-based* explanation (per-prediction feature attribution). Neither alone is sufficient for a non-expert audience.

Relationship-based access control - Beyond conventional RBAC, access to health records is conditioned on an active care relationship between practitioner and patient, established through appointment or explicit owner consent reflecting least-privilege and data-ownership principles from health informatics.

2.4 Existing Technologies and Systems

Table 2 : Comparison of Existing Pet Health Applications vs Proposed System

Feature / Aspect	Existing Pet Health Applications	Proposed System
Symptom handling	Manual symptom checklists; limited behavioral analysis	Structured symptom and behavioral pattern analysis with intelligent interpretation
Disease risk prediction	Mostly descriptive information; no prediction	Machine learning-based early disease risk prediction
Use of AI	Basic rule-based or static logic	Agentic AI with autonomous reasoning, prioritization, and action triggers
Preventive care	Reminders only	Intelligent preventive care agent (vaccination tracking, risk-aware reminders)
Decision support	General information only	Action-oriented support (risk alerts, recommendations, urgency assessment)
Behavior monitoring	Usually missing or very limited	Multi-factor behavior and symptom pattern analysis
User guidance	Generic advice	Personalized guidance based on risk level, pet profile, and detected patterns

System intelligence	Static; no autonomous action	Agentic layer generating alerts and recommendations without user prompts
Data sources	Single-type inputs	Hybrid: symptoms, behaviors, demographics, preventive health data
Veterinary support	Manual clinic search or static lists	Automated location-based discovery, surgeon availability, real-time slots
Explainability	Absent	Ontology-linked reasoning plus model feature attribution
Language support	Typically, English only	Trilingual (English / Sinhala / Tamil)

2.5 Research Gap

The reviewed literature establishes four consistent boundaries:

1. **Prediction without action** - Studies [16], [17], [18], [19] demonstrate predictive capability but terminate at the prediction. None autonomously assess urgency, generate preventive-care actions, or route the owner to appropriate care.
2. **Clinical data dependency** - Work such as [20] and [21] relies on clinical records or laboratory data, which by definition exist only after the owner has sought care, too late for the early-detection objective.
3. **Explainability treated as optional** - Surveys [22], [23] identify trust and reliability as adoption barriers, yet reviewed systems rarely surface *why* a prediction was made in terms a non-expert can act upon.[24], [25], [26]
4. **Absence of care-pathway integration** - No reviewed system connects risk assessment to preventive scheduling and real-time veterinary availability within a single pathway.

Research Gap - No existing system combines machine learning risk prediction from owner-observable data with an agentic reasoning layer that explains its output, assesses urgency, tracks preventive care, and routes the owner to a specific available veterinary professional, while enforcing privacy-preserving, relationship-based access to the resulting health records.

2.6 Conceptual Framework

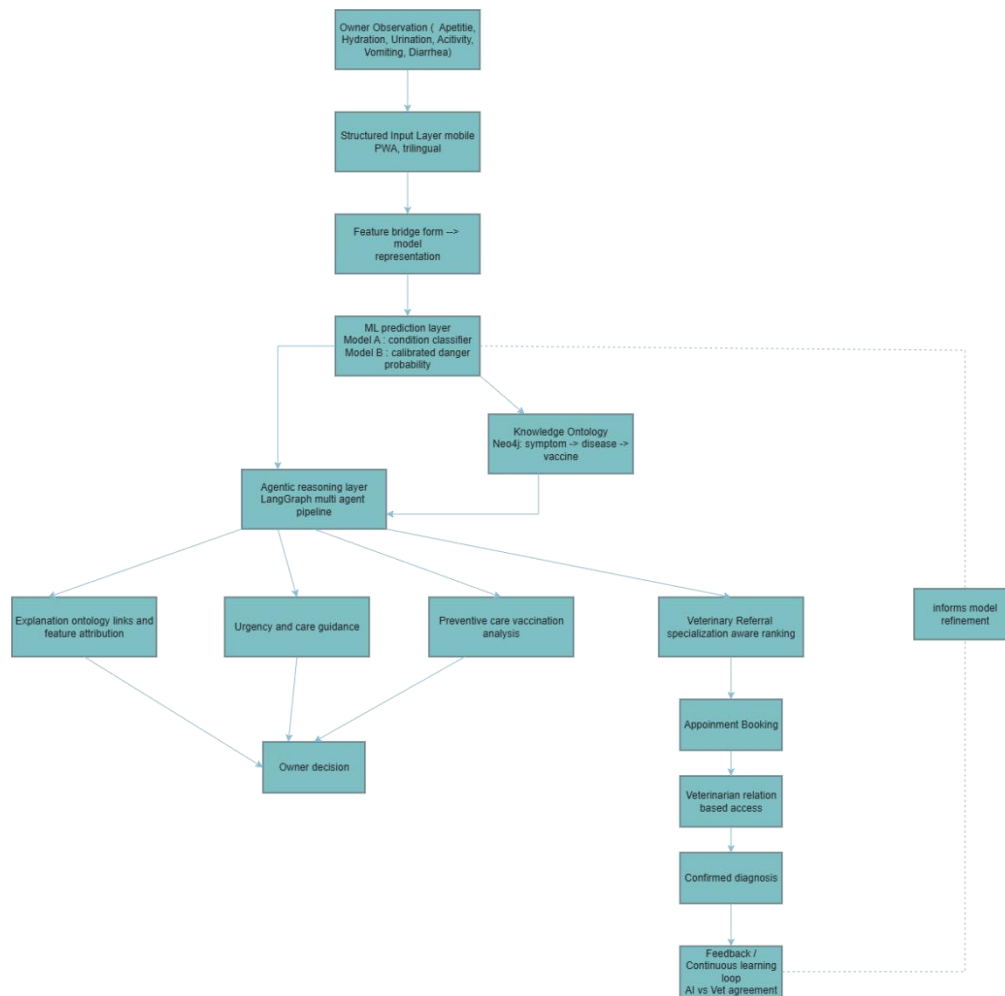


Figure 1 : Conceptual Framework of the Proposed System

The framework is deliberately cyclical. The veterinarian's confirmed diagnosis returns to the system as ground truth, enabling measurement of real-world agreement, a signal distinct from offline test accuracy.

2.7 Chapter Summary

The literature confirms that machine learning can detect early disease signals in companion animals, and that decision support rather than autonomous diagnosis is the appropriate near-term application. It also confirms that current systems stop at prediction, depend on clinical data unavailable at the point of early detection, treat explainability as optional, and do not integrate risk assessment with the wider care pathway. This project addresses that combined gap through an agentic, explainable, care-integrated system operating on owner-observable data.

Chapter 3: Research Methodology

3.1 Introduction

This chapter presents the research design, approach, data strategy, analytical methods, technologies, and ethical framework governing the study. The methodology follows the Design Science Research paradigm, in which the design, construction, and evaluation of a working artefact constitute the primary contribution.

3.2 Research Design and System Diagrams

3.2.1 Use Case Diagram



Figure 2 : Use case Diagram

3.2.2 Entity Relationship Diagram

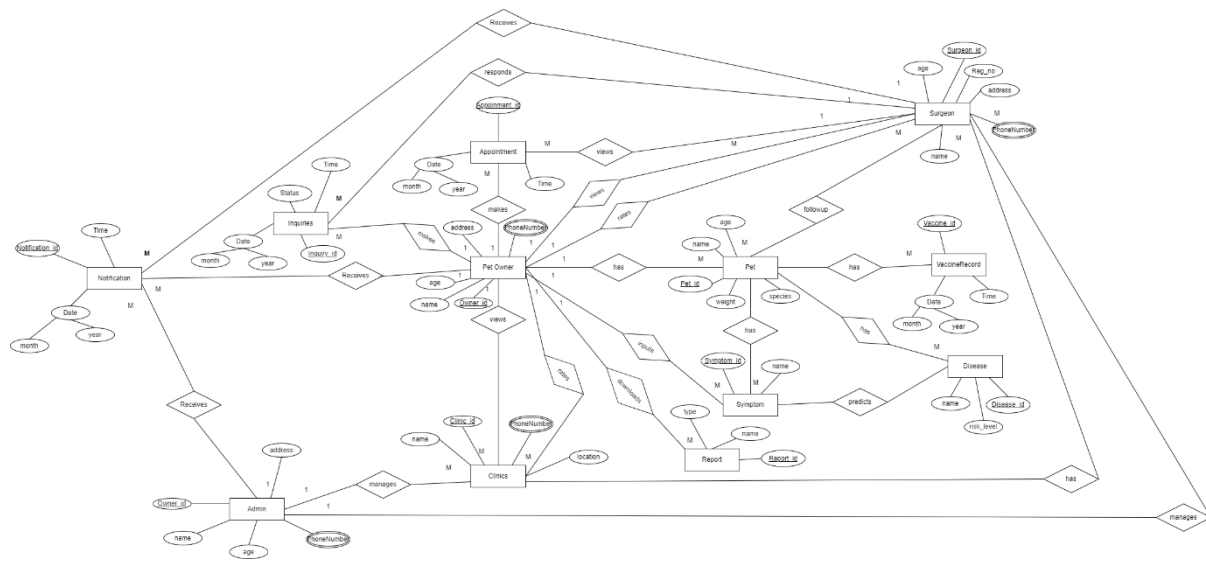


Figure 3: ER Diagram

3.2.3 Activity Diagram

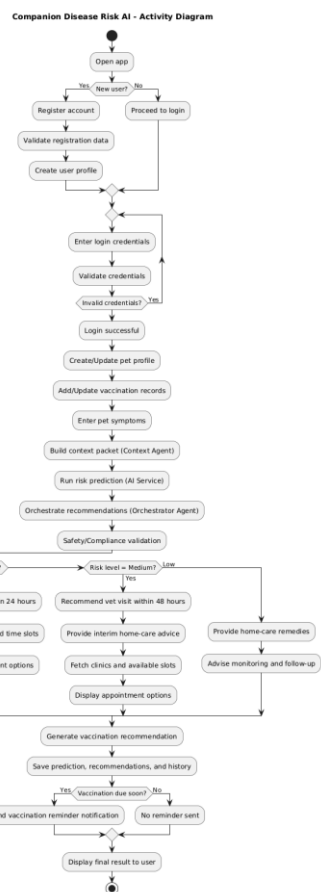


Figure 4: Activity Diagram

3.2.4 Class Diagram

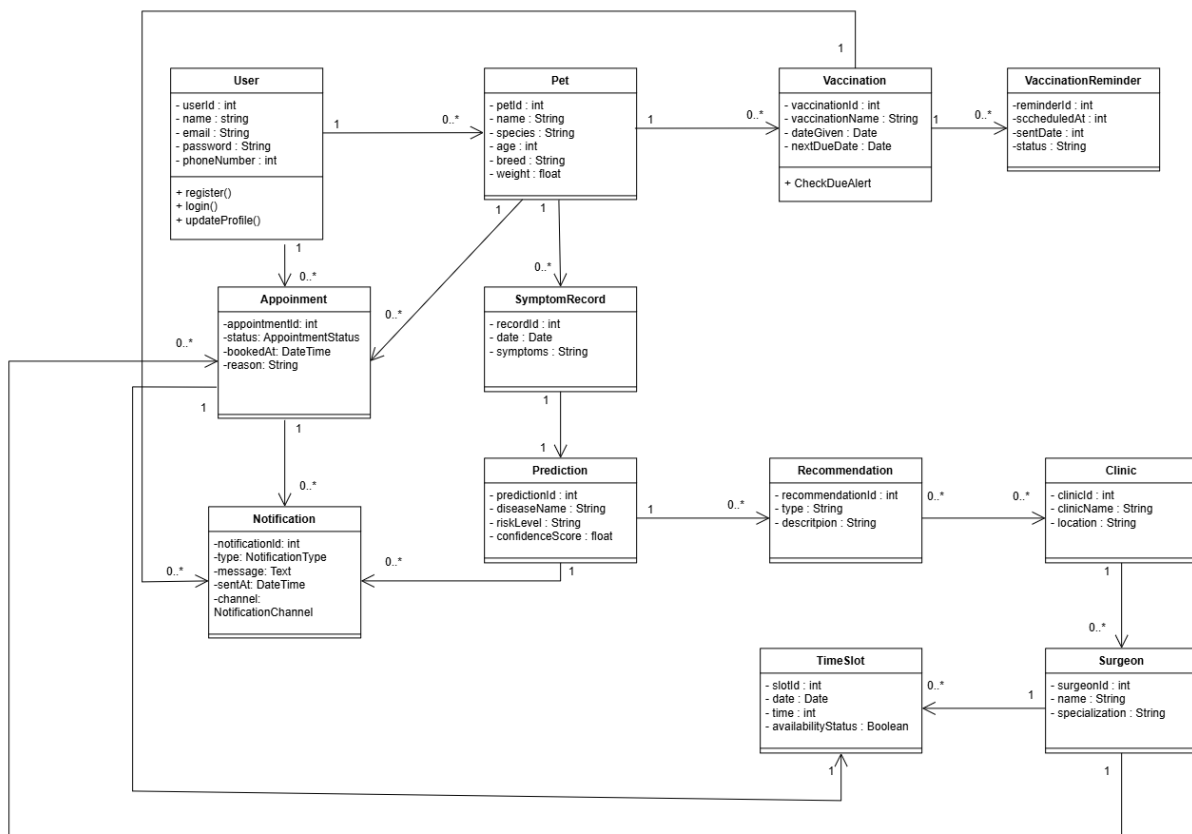


Figure 5: Class Diagram

3.2.5 System Architecture

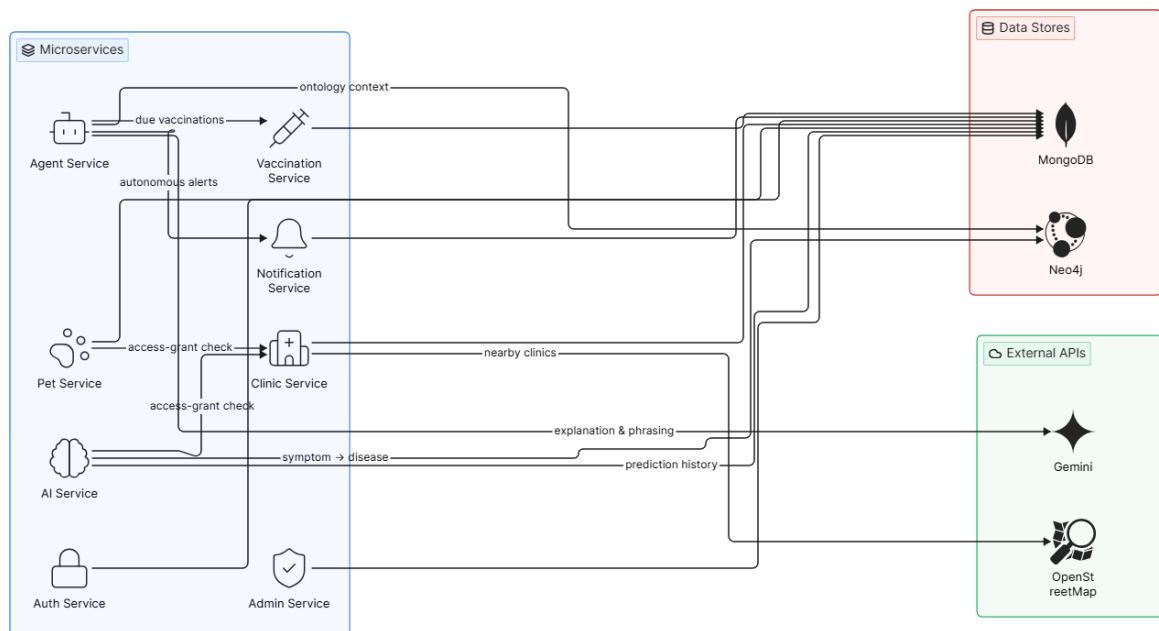


Figure 6: System Architecture

3.3 Research Approach

The study follows a Design Science Research approach, as the primary objective is to design, develop, and evaluate an artefact. A mixed-method strategy is adopted:

- Quantitative methods train and evaluate machine learning models on structured symptom and behavioral data, reporting accuracy, macro-F1, ROC-AUC, PR-AUC, and calibration behavior.
- Qualitative methods assess usability, trust, and practical effectiveness through user feedback and expert review.

System development follows an Agile methodology, permitting continuous refinement in response to testing and feedback. Each iteration produces a working increment which is evaluated before the next design cycle begins.

3.4 Data Collection Methods

Primary data - Model training to date has used validated public datasets supplemented by knowledge-grounded synthetic generation.

Table 3: Datasets Used for Model Training

Dataset	Source	Size	Used For
Pet Health Symptoms	Hugging Face karenwky/pet-health-symptoms-dataset	2,000 rows	Model A - 5 everyday condition classes
Template augmentation	Generated via the runtime feature bridge	500 rows	Model A - closing the train/serve distribution gap
Knowledge-grounded chronic synthesis	Generated from documented clinical sign profiles	2,700 rows	Model A - 6 chronic disease classes
Animal Condition Classification	Kaggle willianoliveiragibin/animal-condition	~871 rows	Model B - danger probability

Secondary data - A structured online survey of dog and cat owners has been administered, capturing observable symptoms (appetite, water intake, urination frequency, activity level, vomiting), vaccination status, health-seeking behavior, and basic pet demographics. Collection is ongoing: 31 valid responses have been received to date against a target of 100–150. The current sample is therefore treated as a *preliminary* dataset suitable for descriptive characterization and for validating the realism of the symptom vocabulary, but not yet sufficient for inferential analysis or for model retraining. Survey data does not inform the machine learning results reported in Chapter 4; the relationship between the two is discussed in Section 4.2.7.

3.5 Sampling Technique and Sample Size

Survey sampling (in progress) - non-probability purposive sampling targeting dog and cat owners in Sri Lanka, distributed through veterinary clinics, animal welfare organizations, and online pet-owner communities. Inclusion criterion: current owner of at least one dog or cat.

A target of 100-150 valid responses was set as sufficient for descriptive characterization of symptom prevalence and for validating the realism of the symptom vocabulary. 31 valid responses (approximately 21-31% of target) have been collected at the time of this report, and collection continues.

The implications of the current sample size are stated plainly rather than minimized:

- The sample supports descriptive reporting of symptom prevalence and health-seeking behavior, and is adequate for face validation of the symptom vocabulary, confirming that the attributes the system asks about are one's owners can actually observe and report.
- It is not sufficient for inferential statistics, subgroup comparison (for example dog versus cat owners), or model retraining, where a sample of this size would risk overfitting to idiosyncratic responses.
- Purposive, self-selecting online recruitment carries a self-selection bias toward more engaged and digitally literate owners, who are plausibly more attentive to their animals' health than the general population. This limits generalizability and is reported as a limitation rather than corrected for.

Continued collection is the highest-priority remaining activity (Section 5.1, A2).

Expert review sampling (planned) - Purposive sampling of 3-5 practicing veterinary surgeons to review a stratified sample of system predictions. A prepared review instrument and answer key already exist (docs/ml/expert_review_sample.csv).

Model evaluation sampling (completed) - Stratified 80/20 train-test split. Model A: 4,160 training / 1,040 test instances. Model B: 651 training / 218 test instances.

3.6 Data Analysis Methods

1. **Classification performance** - Accuracy, macro-F1, and per-class precision/recall/F1, with a confusion matrix. Macro-averaging is reported because class support is deliberately balanced across chronic classes and would otherwise mask minority-class weakness.
2. **Probability quality** - ROC-AUC and PR-AUC for the danger model, interpreted alongside prevalence, plus balanced accuracy to expose degenerate majority-class behavior.
3. **Calibration analysis** - Sensitivity analysis of the base-rate correction, quantifying the trade-off between recovering "low risk" outputs on benign cases and retaining escalation on serious cases.

4. **Distributional validation** - Comparison of the rendered training text distribution against serve-time input to confirm the train/serve gap is closed.
5. **Knowledge-consistency validation** - Verification that predictions for symptom sets derived from ontology relationships recover the expected disease, testing agreement between the statistical and knowledge components.
6. **Real-world agreement** - AI-vs-veterinarian agreement rate computed from confirmed diagnoses, a measure distinct from, and more demanding than, offline test accuracy.
7. **Qualitative analysis** - Thematic analysis of usability feedback and expert commentary.

3.7 Tools and Technologies Used

Table 4: Tools and Technologies

Layer	Technology	Rationale
Front end	React 18, Vite, TypeScript, Tailwind CSS	Component model, fast builds, type safety
Mobile delivery	Progressive Web App (vite-plugin-pwa)	Installable and offline-tolerant without app-store dependency (NFR5)
Mapping	Leaflet + OpenStreetMap	Open, licence-free geospatial data
State / forms	Zustand, react-hook-form, Zod	Lightweight state; schema-validated input (FR2.4)
Backend (Node)	Node.js 20, Express, TypeScript	Mature ecosystem, shared language with front end
Backend (Python)	FastAPI, Python 3.11	Native access to the scientific Python stack
Machine learning	scikit-learn 1.5, NumPy, pandas	Lightweight, interpretable, CPU-only inference (NFR4)
Agentic layer	LangGraph	Explicit graph orchestration with reproducible routing
Language model	Google Gemini (gemini-flash-latest)	Natural-language phrasing, with deterministic rule fallback
Scheduling	APScheduler	Background autonomous monitoring (FR4.4)
Databases	MongoDB 7, Neo4j 5.18 Community	Document store for records; property graph for the ontology
Authentication	JSON Web Tokens, bcrypt	Stateless auth with role claims (FR1.1, NFR2)
Orchestration	Docker, Docker Compose	Reproducible multi-service environment
Monorepo tooling	Turborepo, npm workspaces	Coordinated builds across services
Testing	pytest, Vitest, TypeScript compiler	Unit testing and static verification
Version control	Git / GitHub	Change history and traceability

3.8 Ethical Considerations

The study involves animal health information supplied by human participants and is designed to minimize risk to both.

- **Non-diagnostic positioning (NFR1)** - The system states persistently that it does not replace professional veterinary diagnosis. A disclaimer appears on every risk result, and no interface copy, agent output, or documentation presents results as a diagnosis.
- **Informed and voluntary participation** - Survey participation will be voluntary, with the purpose, data use, and withdrawal rights disclosed before consent.
- **Data minimization and anonymity** - No sensitive personal information is collected. Research data is anonymized; analysis concerns pet health attributes, not identifiable owner characteristics.
- **Security** - Authentication uses bcrypt-hashed credentials and signed JWTs. The gateway verifies every token and stamp's identity header derived from the verified token, preventing client-supplied header spoofing.
- **Privacy by design** - Access to an animal's health records requires an active care relationship, established by appointment or explicit owner consent, and revocable by the owner at any time (Section 4.1.5).
- **Safety of guidance** - Where elevated risk is detected, the system directs the owner toward timely veterinary consultation rather than offering self-treatment guidance.
- **Environmental responsibility** - Lightweight models minimize computational and energy cost; broad device compatibility via PWA reduces hardware-upgrade pressure and electronic waste.

3.9 Chapter Summary

The study adopts Design Science Research with a mixed-method strategy and an iterative SDLC. System design is expressed through use case, entity relationship, data flow, class, and deployment models. Model evaluation to date rests on public and knowledge-grounded synthetic datasets. Primary survey collection is under way with 31 responses received against a 100-150 target, and expert veterinary review is scheduled in the remaining work plan. Ethical safeguards center on non-diagnostic positioning, data minimization, and relationship-based access control.

Chapter 4: Progress to Date

4.1 Work Completed

4.1.1 Literature Review

A comprehensive review of sixteen peer-reviewed sources was completed and approved at the proposal stage, spanning quantitative health scoring, clinical visit classification, sensor fusion, conversational AI, biomarker discovery, and systematic reviews of AI in veterinary medicine. The review established the research gap presented in Section 2.5 and produced the comparative analysis in Table 1. This work is complete.

4.1.2 Requirement Gathering

Functional and non-functional requirements were formalized from the approved proposal into a structured specification comprising eight functional requirement groups (FR1-FR8) and seven non-functional requirements (NFR1-NFR7). Each requirement is traceable to a proposal objective. This work is complete; requirements have been treated as a stable baseline throughout implementation.

4.1.3 System Analysis

System analysis produced the use case model (Figure 2), identifying three actors pet owner, veterinarian, administrator and sixteen use cases. Analysis of data ownership across these actors identified a requirement not present in the original proposal: veterinarians must not have unrestricted access to all animal records. This finding drove the access control design in Section 4.1.5.

4.1.4 System Design

The system was designed as a **microservice architecture** behind an API gateway, with a React PWA client. Nine services are implemented and operational.

Table 5: Microservice Inventory

Service	Port	Responsibility
api-gateway	4000	JWT verification, rate limiting, identity-header stamping, request routing
auth-service	4001	Registration, login, JWT issuance, role normalization, admin-provisioned vet accounts
pet-service	4002	Pet profile CRUD, ownership enforcement, grant-scoped veterinary access
clinic-service	4003	Clinics, surgeons, time slots, appointments, booking conflict checks, access grants

notification-service	4004	In-app notifications with deduplication idempotency
vaccination-service	4005	Vaccination records, due and overdue queries
admin-service	4006	Registration approvals, support tickets, append-only audit trail
ai-service	8001	Machine learning inference, prediction persistence, ontology application
agent-service	8002	LangGraph agent pipeline, autonomous monitoring agent

Design decisions of note:

- **Gateway-enforced identity** - Services never trust client-supplied identity. The gateway strips and re-stamps x-user-id and x-user-role from the verified token, making header spoofing structurally impossible.
- **Shared feature bridge** - A single module renders the structured form into the model’s input representation, used identically at training-augmentation time and at serve time, eliminating the train/serve distribution gap.
- **Graph-orchestrated agents** - Agent routing uses deterministic edges with a conditional branch, rather than open-ended autonomy, ensuring reproducible behavior for evaluation.

4.1.5 Prototype Development

A working end-to-end prototype has been implemented and is operational across all nine services with both databases.

Machine learning layer - Two complementary models are blended with a deterministic severity index. Model A classifies the reported presentation into one of eleven conditions. Model B produces a calibrated probability that the presentation is dangerous. Full results appear in Section 4.2.

Agentic layer - A LangGraph pipeline orchestrates three agents plus a composer, with a fourth agent running autonomously in the background.

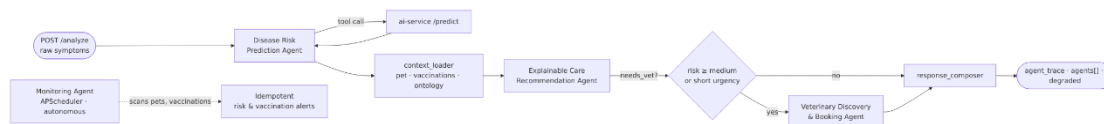


Figure 7: Agentic AI Pipeline (LangGraph Orchestration)

Every response carries an agent_trace, a per-agent breakdown, and a degraded flag. Where the language model is unavailable or rate-limited, the entire layer degrades to deterministic rules and the interface displays a “Rule-based mode” badge, so the system never hard-fails.

Explainability - Two mechanisms are surfaced. Knowledge-based explanation presents ontology relationships between reported symptoms and candidate conditions. Model-based explanation presents per-prediction feature attributions, aggregated back onto the form fields the owner completed (Section 4.5).

Feedback and continuous learning loop - Owners rate each assessment as helpful or not helpful and indicate whether it matched their veterinarian's diagnosis. Veterinarians record the confirmed actual diagnosis. Both are stored on the prediction document, producing a clean prediction-versus-actual join from which an AI-vs-veterinarian agreement rate is computed.

Relationship-based access control - Analysis identified that veterinarians could initially read every pet record and every assessment in the system. This was redesigned around an explicit care relationship.

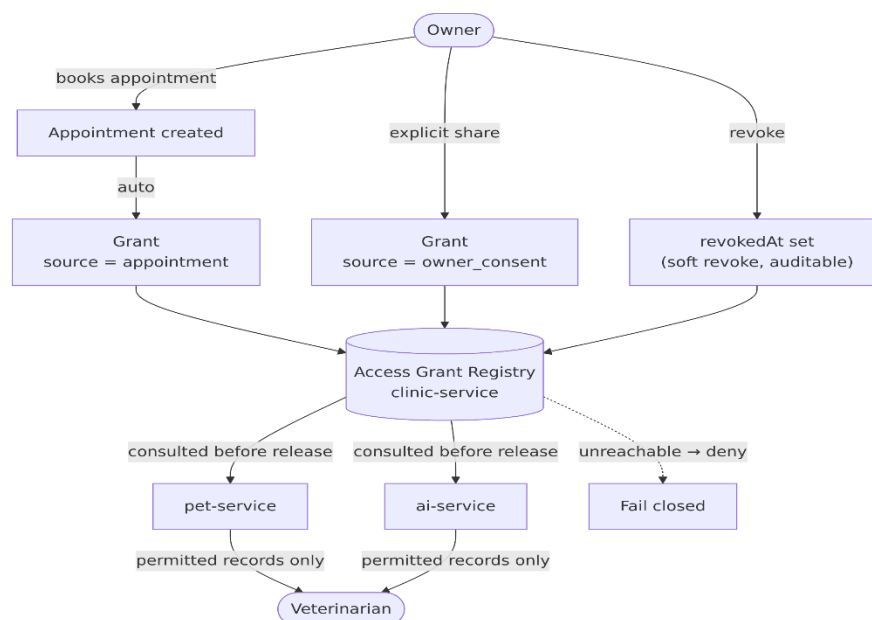


Figure 8: Relationship-Based Access Control Model

The rule enforced is: a veterinarian may read a pet's health information only while an active grant links them to that pet. Grants arise from an appointment with that veterinarian persisting beyond the visit to provide continuity of care for returning patients or from explicit owner consent, revocable at any time. Revocation is soft, retaining the record so that access history remains auditable. Enforcement resides in the services rather than the interface, and fails closed: if the grant registry is unreachable, access is denied rather than granted.

4.2 Current Findings

4.2.1 Model A - Condition Classifier

Pipeline - TF-IDF vectorization (1-2 grams) followed by logistic regression with balanced class weighting. Split. 4,160 training / 1,040 test instances, stratified. Headline result. Accuracy 0.903, macro-F1 0.908.

Table 6: Model A Per-Class Performance

Class	Precision	Recall	F1	Support	Data Origin
Diabetes Mellitus	1.000	1.000	1.000	90	Synthetic (knowledge-grounded)
Heart Disease	1.000	1.000	1.000	90	Synthetic
Liver Disease	1.000	1.000	1.000	90	Synthetic
Pancreatitis	1.000	1.000	1.000	90	Synthetic
Chronic Kidney Disease	0.947	1.000	0.973	90	Synthetic
Cancer	1.000	0.944	0.971	90	Synthetic
Digestive Issues	0.890	0.848	0.868	105	Real dataset
Mobility Problems	0.748	0.883	0.810	94	Real dataset
Ear Infections	0.777	0.816	0.796	98	Real dataset
Parasites	0.831	0.763	0.796	97	Real dataset
Skin Irritations	0.804	0.736	0.768	106	Real dataset
Macro average	0.909	0.908	0.908	1,040	-

Interpretation - stated honestly - The headline figure is inflated by the six chronic classes, whose F1 scores of 0.971-1.000 reflect the separability of synthetically encoded clinical patterns, not diagnostic accuracy on real cases. The five classes derived from real data sit at F1 0.768-0.868 and confuse with one another, particularly Skin Irritations with Ear Infections and Parasites a clinically plausible confusion given overlapping presentation. The per-class breakdown, not the headline accuracy, is the appropriate figure to quote.

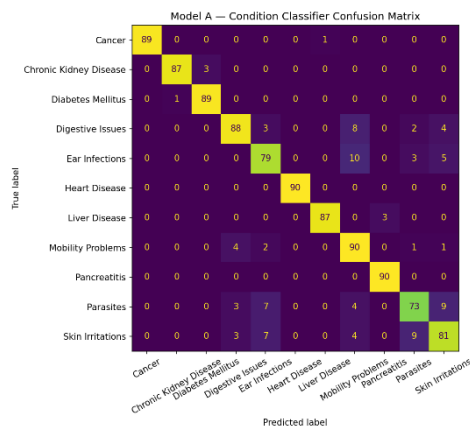


Figure 9: Model A Normalized Confusion Matrix

4.2.2 Model B - Danger Probability

Pipeline - multi-label binarization of symptom tokens, then logistic regression with balanced weighting wrapped in a calibrated classifier (sigmoid). **Split** - 651 training / 218 test instances.

Table 7: Model B Performance and Imbalance Caveat

Metric	Value	Interpretation
ROC-AUC	0.964	The probability ranks dangerous above non-dangerous cases well
PR-AUC	0.999	Inflated by prevalence not evidence of quality
Positive rate (training)	0.977	Severe class imbalance
Balanced accuracy	0.500	Degenerate at the 0.5 threshold
Recall (dangerous)	1.000	Predicts the majority class
Recall (not dangerous)	0.000	Fails entirely on the minority class

Interpretation and mitigation - A dataset that is 97.7% “dangerous” means a constant classifier achieves 97% accuracy while learning nothing. Model B is therefore never used as a hard label. It contributes only a calibrated probability, which is first prior-corrected shifting from the training prior of 0.977 to a neutral deployment prior of 0.5 and then blended with a deterministic severity index derived from the reported vitals. Calibration sensitivity analysis showed this correction recovered approximately 59% “low risk” outputs on benign cases while retaining roughly 99% of serious chronic cases at medium risk or above, correcting a systematic over-triage tendency. This is reported as a documented limitation and a deliberate design response, not as a strength.

4.2.3 Knowledge Ontology

Table 8: Knowledge Ontology Composition

Element	Count
Disease nodes	20
Symptom nodes	24
Vaccine / preventive nodes	8
INDICATES relationships (symptom → disease, weighted)	48
PREVENTS relationships (vaccine → disease)	12

Every node carries Sinhala and Tamil name properties alongside English. The schema is applied automatically and idempotently at ai-service startup. Symptom node names are maintained in exact correspondence with the model’s checklist tokens and the front-end symptom identifiers; divergence between these three would silently degrade prediction quality rather than raise an error, so this correspondence is treated as an invariant.

4.2.4 Agentic Layer

Verified behavior on a representative case (lethargy, anorexia, persistent vomiting, moderate diarrhea, four days’ duration):

- All three pipeline agents executed with degraded: false, confirming live language-model operation.
- agent_trace recorded the full path: disease risk prediction → context loader → explainable care recommendation → veterinary discovery → response composer.
- Required specialization was correctly derived as emergency care from the predicted condition.
- Clinic ranking selected a recommended clinic and identified the matching surgeon with their next free slot.
- With the language model unavailable, the identical request returned rule-based output with degraded: true, confirming graceful degradation.

The autonomous monitoring agent was verified to scan pets and vaccination records, create risk and vaccination notifications, and correctly suppress duplicates on a second run within the same day.

4.2.5 Requirements Status

Table 9: Requirements Implementation Status

Requirement	Status	Evidence
FR1 - User & pet profile management	Complete	Three roles; ownership enforced from verified identity
FR2 - Symptom & behavior collection	Complete	Structured vitals + checklist + duration + notes
FR3 - ML risk prediction	Complete	Two models; metrics in Tables 6 and 7

FR4 - Agentic AI layer	Complete	Four agents; trace, per-agent breakdown, degraded flag
FR5 - Vaccination management	Complete	Persisted records; due/overdue queries; risk-aware escalation
FR6 - Veterinary discovery & booking	Complete	Haversine ranking, live OpenStreetMap results, slot booking with conflict checks
FR7 - Alerts & notifications	Partial	In-app notifications complete; push notification (FCM) out of scope
FR8 - Admin & vet functions	Complete	Clinic/surgeon/slot management; vet provisioning; learning metrics panel
NFR1 - Decision support, not diagnosis	Complete	Disclaimer on every result; verified across all copy
NFR2 - Security & privacy	Complete	JWT + bcrypt; gateway identity stamping; relationship-based access control
NFR3 - Usability	Partial	Mobile-first trilingual UI built; formal usability testing pending
NFR4 - Performance / efficiency	Complete	Model artefacts under 0.5 MB; CPU-only inference
NFR5 - Compatibility	Complete	Installable PWA
NFR6 - Reliability & maintenance	Complete	Automated tests; health checks; fail-closed authorization
NFR7 - Evaluation	Partial	Model metrics complete; usability and expert evaluation pending

4.2.6 Verification and Testing

Table 10: Test Coverage Summary

Suite	Tests	Coverage
ai-service (pytest)	19	Feature bridge mappings, prediction with models and fallback, history persistence
agent-service (pytest)	15	Rule ladder, degraded operation, graph routing, safety floor

api-gateway (Vitest)	7	JWT verification, public allowlist, enforcement on/off, header injection
Static verification	-	TypeScript strict-mode compilation across all workspaces

Access control was verified through an explicit test matrix confirming that an ungranted veterinarian receives zero records on both list and direct-fetch endpoints; that a granted veterinarian receives exactly the permitted pet; that a different veterinarian remains denied throughout; that revocation immediately removes access; and that the owner's own view is unaffected in every case. Recording a diagnosis was confirmed to require an active grant.

4.2.7 Preliminary Survey Findings

The owner survey described in Section 3.4 has returned 31 valid responses, all with recorded informed consent. Collection remains open. Analysis at this stage is descriptive; the sample is below the 100-150 target and no inferential claims are drawn from it.

Table 11: Respondent and Animal Characteristics (n = 31)

Attribute	Category	n	%
Species	Dog	22	71.0
	Cat	9	29.0
Sex	Male	18	58.1
	Female	13	41.9
Age band	Under 1 year	1	3.3
	1-3 years	10	33.3
	4-7 years	16	53.3
	8 years and over	3	10.0
Consent	Given	31	100.0

Age was reported as free text and parsed for 30 of 31 responses (mean 4.11 years, median 4.0, range 0.7-10.0). One respondent recorded “No idea, he was old when I adopted him from nowhere” a rescue animal of unknown age. This is a meaningful design signal rather than a nuisance value: the system must accept an unknown age without blocking assessment, since a non-trivial proportion of Sri Lankan companion animals are adopted strays.

Table 12: Distribution of Owner-Reported Health Indicators (n = 31)

Indicator	Response	n	%
Appetite	Normal	21	67.7
	Slightly reduced	10	32.3

Water intake	Normal	20	64.5
	Increased	6	19.4
	Decreased	5	16.1
Activity level	Normal	17	54.8
	Slightly less active	10	32.3
	Very low activity	4	12.9
Vomiting	Occasional	21	67.7
	No	10	32.3
Diarrhea (<i>item answered by 12 of 31</i>)	No	7	22.6
	Occasional	5	16.1
	Not answered	19	61.3
Veterinary diagnosis	Not diagnosed / not sure	24	77.4
	Chronic Kidney Disease	4	12.9
	Cancer	1	3.2
	Arthritis	1	3.2
	Injection allergies	1	3.2

Findings

F1 - The symptom vocabulary is answerable, every respondent completed every core indicator (appetite, water intake, activity, vomiting) using the supplied categories, with no free-text clarification requested. This satisfies the survey’s primary purpose: confirming that the attributes the system asks about are one’s owners can observe and report. The vocabulary requires no revision.

F2 - Reported abnormality is high, and this must be interpreted with care. Treating “Normal”/ “No” as unremarkable, 26 of 31 respondents (83.9%) reported at least one abnormal indicator, and 8 (25.8%) reported three or more. Two readings are possible and both carry consequences:

- Self-selection. Owners with a currently unwell animal are more motivated to complete a pet-health survey. This is the more likely explanation and reinforces the sampling limitation stated in Section 3.5.
- Category sensitivity. “Occasional” vomiting reported by 67.7% is common in clinically healthy cats and dogs, particularly in relation to grooming or diet. If owners read “Occasional” as encompassing entirely normal variation, the category may over-capture.

The second reading has a direct design implication: an input vocabulary that labels benign variation as abnormal will bias the severity index upward and contribute to over-triage. This finding therefore corroborates, from an independent direction, the over-triage tendency already identified statistically in Model B (Section 4.2.2) and mitigated through prior correction.

F3 - Ranking of reported abnormalities. Vomiting was the most frequently abnormal indicator (67.7%), followed by activity level (45.2%), water intake (35.5%), appetite (32.3%), and diarrhea (16.1% of the full sample). The prominence of activity and appetite changes supports the emphasis these carry in the severity index.

F4 - Response range was narrower than the instrument allowed. No respondent selected the most severe option for appetite (“significantly reduced”) or for vomiting (“frequent”). Observed responses cluster in the normal-to-mild range. A sample containing no severe presentations cannot validate the system’s behavior at the high-risk end of its scale, which remains dependent on synthetic and expert evaluation.

F5 - The diagnosed subgroup is small but analytically instructive. Seven respondents (22.6%) reported a veterinary-confirmed condition, four of them chronic kidney disease. Examining owner-reported indicators against those diagnoses:

Species	Diagnosis	Abnormal indicators reported
Cat	Chronic Kidney Disease	4 - appetite, water intake, activity, vomiting
Dog	Chronic Kidney Disease	4 - appetite, activity, vomiting, diarrhea
Dog	Chronic Kidney Disease	5 - appetite, water intake, activity, vomiting, diarrhea
Dog	Chronic Kidney Disease	0 - all indicators reported normal
Cat	Cancer	4 - appetite, water intake, activity, vomiting
Cat	Injection allergies	3 - water intake, activity, vomiting
Dog	Arthritis	2 - activity, vomiting

Three of the four CKD cases presented multiple abnormal indicators, and the cat case showed increased water intake with very low activity consistent with the polydipsia and lethargy characteristic of the condition, and with the INDICATES weights encoded in the ontology (Section 4.2.3).

The fourth CKD case reported every indicator as normal. This single response is the most valuable observation in the dataset. It is a direct, real-world instance of the central limitation of owner-observable data: a confirmed chronic condition that presents no signal the owner can perceive. No amount of model refinement can detect what the input channel does not carry. This is precisely why the system is positioned as decision support rather than diagnosis, why the disclaimer requirement (NFR1) is non-negotiable, and why a “low risk” output must never

be presented as reassurance that the animal is well. The final report will cite this case explicitly when discussing limitations.

F6 - Instrument defect identified. The diarrhea item was answered by only 12 of 31 respondents (38.7%). Inspection of response timestamps shows all 19 missing values occur in responses submitted before 10 January 2026 22:02, and every response after that point contains an answer. This indicates the item was added or made visible partway through collection rather than being skipped by respondents. Consequently, diarrhea prevalence cannot be reported for the full sample and the 16.1% figure above is computed over the whole sample rather than over those who saw the question (5 of 12, or 41.7%, among respondents who were asked).

Role of the survey within the research design

It is worth stating precisely what this survey does and does not contribute, since its role is easily misread:

1. Face validation of the symptom vocabulary (primary purpose). Achieved see F1.
2. Empirical support for the problem statement. Partially achieved. The 22.6% diagnosed rate and the CKD case with no observable signs support the premise that early disease is difficult for owners to detect. However, the deployed instrument does not ask how long owners waited before consulting a veterinarian, so health-seeking delay a central claim in Section 1.2 remains unquantified by primary data.
3. Language and adoption preferences. Not captured. The deployed instrument omits the technology-use and language-preference sections described in the planned design, so the trilingual interface priority remains justified by context rather than by respondent data.
4. What it does *not* do. The survey is not a training data source. Owner-reported symptom sets carry no veterinary-confirmed label for the majority of responses, and the seven that do are far too few to supervise an eleven-class classifier. Model training rests on the datasets in Table 3; the route to label-bearing real-world data is the feedback loop in Section 4.1.5.

Points 2 and 3 identify a gap between the planned and deployed instruments. Extending the questionnaire with health-seeking delay, vaccination awareness, and language preference items is recorded as Activity A2b in Section 5.1, to be applied to the remaining collection.

This distinction matters for the evaluation narrative: a larger survey improves the *justification* and *usability grounding* of the system, whereas improved *predictive validity* depends on expert review (Activity A4) and on accumulated veterinarian-confirmed diagnoses.

4.3 Challenges Encountered

C1 - Severe class imbalance in the danger dataset. The Kaggle dataset is 97.7% positive, producing a model with zero minority-class recall and a systematic tendency to over-triage benign presentations as dangerous.

C2 - Train/serve distribution gap. The application collects structured form data, whereas the source dataset consists of free text. A model trained on one and served the other would degrade silently.

C3 - Fragmented and unreadable feature attribution. TF-IDF over a rendered sentence decomposes a single owner answer into several overlapping n-grams. One answer “drinking less water” surfaced as four separate entries (“less”, “drinking less”, “less water”, “water is”) plus template connectives such as “has”. This misrepresented one finding as four independent pieces of evidence and was unintelligible to a non-expert.

C4 - Service startup race condition. The Neo4j container requires approximately 70 seconds to expose its Bolt interface. The AI service applies the ontology once at startup and was reaching the database before it was ready, leaving the graph silently empty disabling explanation without producing any error.

C5 - Over-permissioned veterinary access. The initial design allowed any authenticated veterinarian to read every pet profile and every AI assessment on the platform, which is inappropriate for health-adjacent data.

C6 - Fabricated data in the interface. Several elements displayed invented values: fixed star ratings and review count for clinics despite no review feature existing; a hardcoded vaccination reminder shown regardless of records; a fixed symptom list displayed when none were reported; and a fabricated confidence value during offline operation.

C7 - Unauthenticated internal requests. Three administrative interface calls omitted the bearer token, so with authentication enforcement enabled they failed silently leaving governance panels blank and preventing delivery of credentials to newly provisioned veterinarians.

4.4 Solutions Implemented

Table 13: Challenges and Solutions

#	Challenge	Solution Implemented	Outcome
C1	Danger-model imbalance	Bayesian base-rate correction from training prior 0.977 to deployment prior 0.5; probability blended with a deterministic severity index; hard label never used	~59% “low risk” recovered on benign cases; ~99% of serious cases retained at $\geq$ medium

C2	Train/serve gap	Single shared feature bridge used identically for training augmentation and serve-time rendering	Distribution gap closed; validated in <code>distributional_validation.json</code>
C3	Fragmented attribution	N-grams mapped back to the originating form field and contributions summed per field (valid for a linear model); unmappable template fragments discarded; stored predictions re-attributed on read	“less / drinking less / less water / water is / has” became “Water intake: increased” and “Appetite: reduced”
C4	Startup race	Container health check probing the Bolt interface and authentication, gating dependent service startup	Ontology now applies reliably on cold start; verified across restarts
C5	Over-permissioned access	Relationship-based access control with appointment and consent grants, enforced in services, failing closed	Ungranted veterinarian access reduced from all records to zero; verified by test matrix
C6	Fabricated data	Ratings removed from schema, seed and interface; demo seeding made opt-in; empty states replace invented values; offline confidence suppressed in both interface and exported reports	No fabricated values remain in any user-facing surface
C7	Unauthenticated calls	Routed through the authenticated API client used elsewhere in the application	Governance panels and credential delivery restored

Two of these solutions constitute methodological contributions rather than mere defect repairs. The attribution aggregation in C3 addresses a general weakness of TF-IDF explainability for form-derived text, and the access model in C5 formalizes a privacy property that the original proposal did not specify.

Chapter 5: Proposed Work Plan

5.1 Remaining Activities

Table 14: Remaining Activities

ID	Activity	Description	Priority
A1	Survey instrument design - complete	Questionnaire designed and deployed; 31 responses received	Done
A2	Continue survey collection	Extend distribution to reach 100-150 responses; currently at 31. Widen channels: veterinary clinics, welfare organizations, university networks, pet-owner groups	High
A2b	Extend the questionnaire	Add the health-seeking delay, vaccination awareness, and language preference items identified as missing in Section 4.2.7; repair the diarrhea item so it is presented to all respondents	High
A3	Survey re-analysis	Re-run descriptive analysis on the enlarged sample; test whether the 83.9% abnormality rate persists or reflects early self-selection	High
A4	Expert veterinary review	Distribute the prepared review sample to 3-5 practicing surgeons; analyze agreement	High
A5	Formal usability testing	Task-based evaluation with representative owners; System Usability Scale administration	High
A6	Model B dataset improvement	Source or construct a better-balanced danger	Medium

		dataset to address the imbalance limitation	
A7	Interface screenshot capture	Systematic capture of all flows for the final report	Medium
A8	Push notification integration	Evaluate feasibility of completing FR7.2 within scope	Low
A9	Performance benchmarking	Inference latency, end-to-end response time, resource consumption	Medium
A10	Final system hardening	Security review, error-handling audit, deployment configuration	Medium
A11	Final report writing	Full dissertation incorporating all evaluation results	High
A12	Viva preparation	Demonstration script, anticipated questions, contribution defence	High

5.2 Timeline (Gantt Chart)

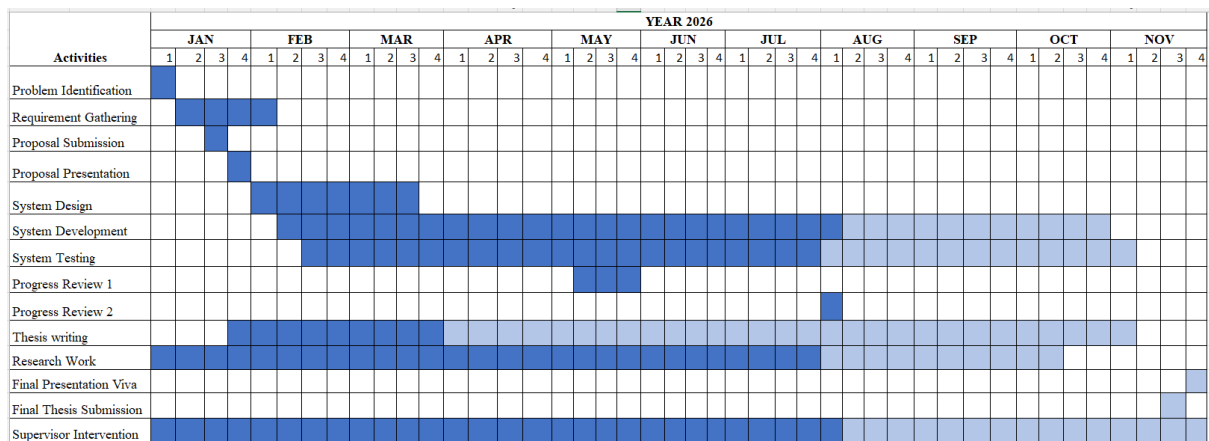


Figure 10: Project Timeline

5.3 Expected Deliverables

1. A fully functional Agentic AI decision support system for companion animal health monitoring, delivered as an installable progressive web application.
2. An agentic symptom and behavior analysis module for early disease risk detection.
3. Machine learning classification models for early risk of selected high-impact diseases, with documented evaluation.

4. An autonomous health-monitoring agent performing continuous evaluation without user prompting.
5. A preventive-care agent tracking vaccination timelines and generating risk-aware alerts.
6. A vaccination reminder system integrated with disease risk analysis.
7. Real-time location-based veterinary clinic discovery with surgeon availability and consultation slots.
8. A risk-based alert and guidance system recommending timely veterinary consultation.
9. Secure user and pet profile management with relationship-based access control.
10. An explainability subsystem combining ontology-based and model-based explanation.
11. A feedback and continuous-learning loop reporting AI-versus-veterinarian agreement.
12. A research report documenting methodology, experimental results, and evaluation.
13. Performance evaluation and usability testing results.
14. System documentation and user guidelines.

5.4 Risk Management Plan

Table 15: Risk Management Plan

ID	Risk	Likelihood	Impact	Mitigation
R1	Insufficient survey responses within the collection window	Medium	Medium	Distribute through multiple channels (clinics, welfare organizations, online communities); the system does not depend on survey data for core function, so a shortfall degrades validation depth rather than blocking delivery
R2	Difficulty recruiting veterinary experts for review	Medium	High	Review instrument and answer key already prepared to minimize expert time; approach supervisors' professional networks; reduce to three reviewers if necessary
R3	No better-balanced danger dataset obtainable	Medium	Medium	Current mitigation (prior correction plus severity blending) is

				already documented and defensible; the limitation would be reported honestly rather than concealed
R4	Language model API quota, cost, or availability limits	Low	Low	Deterministic rule-based fallback already implemented and tested; the system degrades rather than fails
R5	Synthetic training data undermines claims of clinical validity	High	Medium	Already mitigated by reporting per-class results separately and stating the limitation explicitly; expert review (A4) provides independent evidence
R6	Scope expansion during evaluation	Medium	Medium	Requirements baseline frozen; FR7.2 already classified as out of scope; new findings recorded as future work
R7	Data loss or environment failure	Low	High	Source code version-controlled on GitHub; environment fully reproducible via Docker Compose; model artefacts committed
R8	Time pressure from concurrent academic commitments	Medium	High	Evaluation activities scheduled in parallel where independent; final report drafting begins before evaluation concludes

Conclusion

Summary of Progress

The project has progressed from an approved proposal to a fully operational prototype implementing all five research objectives. The literature review, requirements specification, system analysis, and system design are complete. The prototype comprises nine microservices, two machine learning models, a four-agent AI layer, a trilingual progressive web interface, and a knowledge ontology of twenty diseases, twenty-four symptoms, and sixty weighted relationships.

Achievements to Date

1. Working end-to-end system. All five proposal objectives are implemented and demonstrable, from symptom capture through risk prediction, explanation, preventive-care analysis, and veterinary referral to appointment booking.
2. Evaluated machine learning models. Model A achieves accuracy 0.903 and macro-F1 0.908 across eleven classes, with per-class results reported separately for synthetic and real-data classes. Model B's imbalance limitation is quantified, mitigated through base-rate correction, and reported transparently rather than concealed.
3. Genuine agentic behavior. A LangGraph pipeline of three agents with conditional routing, plus a fourth autonomous monitoring agent, verified operating in both live and degraded modes.
4. Explainability by construction. Ontology-linked reasoning and model feature attribution, with attribution redesigned to express itself in terms of the owner's own answers rather than model-internal n-grams.
5. Closed feedback loop. Owner feedback and veterinarian-confirmed diagnoses combine into an AI-versus-veterinarian agreement metric, providing a real-world accuracy signal distinct from offline test performance.
6. Privacy-preserving access control. A relationship-based model restricting veterinary access to animals with an active care relationship, enforced at the service layer and failing closed.
7. Verified integrity of presentation. A systematic audit removed all fabricated data from user-facing surfaces, ensuring that every displayed value derives from real system state.

Next Steps

Immediate priorities are extending survey collection from the current 31 responses toward the 100-150 target, and securing expert veterinary review. These provide the independent validation the project currently lacks: the machine learning results reported here derive from public and knowledge-grounded synthetic datasets, and while the system's behavior is verified, its clinical plausibility has not yet been assessed by practitioners. Formal usability testing will address NFR3 and NFR7, and improving the danger-model dataset remains the most valuable technical refinement available.

The project is on schedule, with the substantial engineering risk now retired and the remaining work concentrated in evaluation and documentation.

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Appendices

Appendix A - Survey Questionnaire

A.1 Instrument as deployed

The following items were administered and produced the 31 responses analyzed in Section 4.2.7.

#	Item	Response options
1	Consent - “This survey is conducted for academic research purposes only. It does not provide medical diagnosis or veterinary advice. I voluntarily agree to participate.”	Yes, I agree
2	Pet type	Dog / Cat
3	Pet age	Free text
4	Pet gender	Male / Female
5	Compared to normal, how was your pet’s appetite?	Normal / Slightly reduced / Significantly reduced
6	Have you noticed any change in your pet’s water consumption?	Normal / Increased / Decreased
7	How would you describe your pet’s activity level?	Normal / Slightly less active / Very low activity
8	Has your pet experienced vomiting?	No / Occasional / Frequent
9	Has your pet experienced diarrhea?	No / Occasional / Frequent
10	Has your pet been diagnosed by a qualified veterinarian with any of the following conditions?	Condition list / Not diagnosed / Not sure

Known issues with the deployed instrument (identified in Section 4.2.7 and addressed by Activity A2b):

- Item 9 (diarrhea) was not presented to the first 19 respondents and is answered by only 12 of 31.

- Item 3 (age) is free text, producing values such as “1year and 6month old” and “No idea” that require manual parsing. A structured age band with an explicit “unknown” option is preferable.
- The instrument does not capture urination frequency, although this is a core system input and a primary indicator for kidney and metabolic conditions.

A.2 Proposed additional items for continued collection

To be added under Activity A2b:

Health-seeking behavior: typical delay between noticing a change and consulting a veterinarian; factors influencing that decision. (Required to substantiate the delay claim in Section 1.2 with primary data.)

Preventive care: vaccination status; awareness of the next due date; current reminder practices.

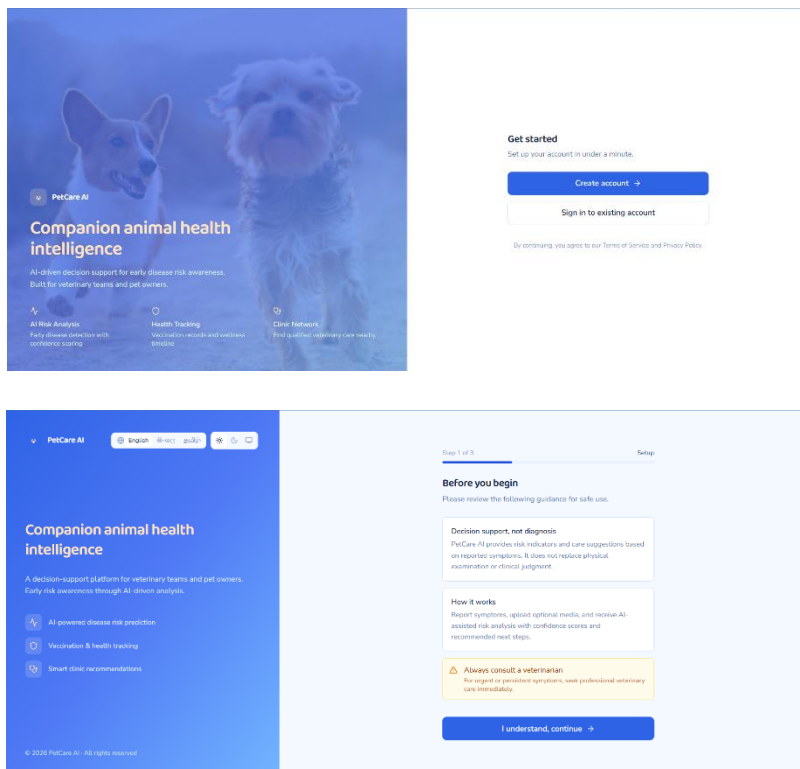
Technology use: existing use of any pet health application; barriers to adoption.

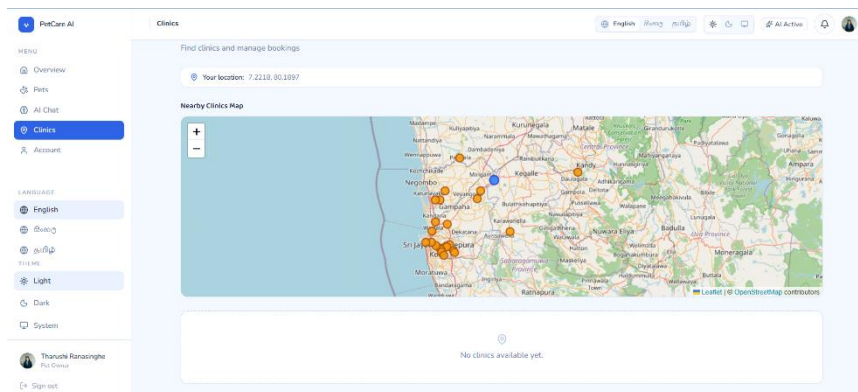
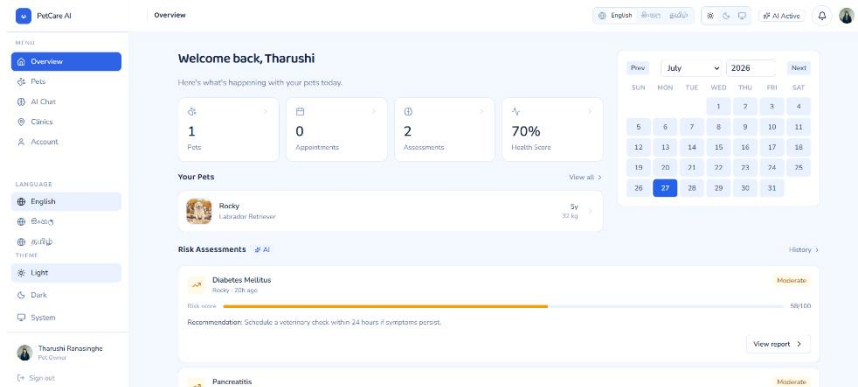
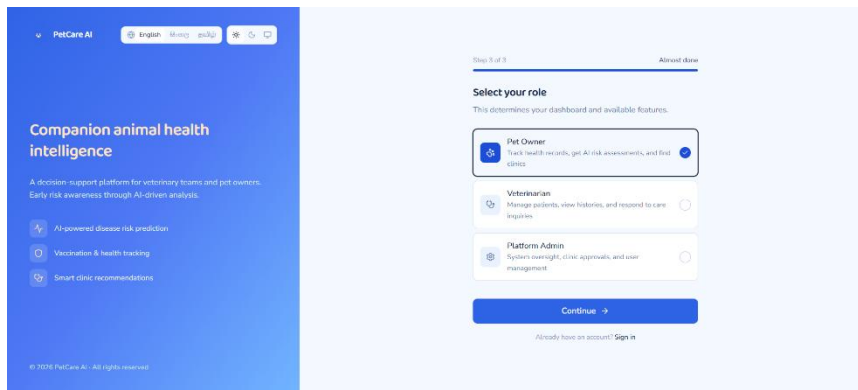
Language preference: English / Sinhala / Tamil. (Required to substantiate the trilingual design decision with respondent data.)

Urination frequency: normal / increased / reduced, to align the instrument with the system’s actual input set.

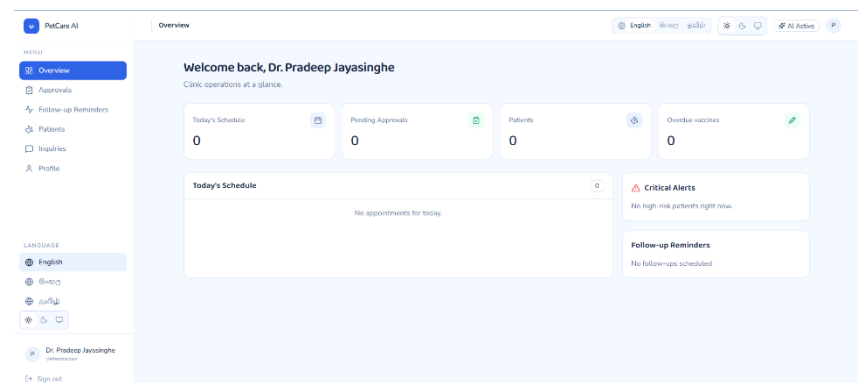
Appendix B - Screenshots

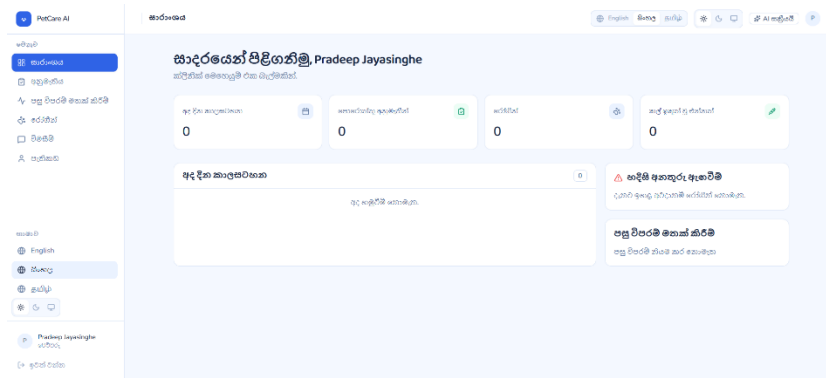
WEB – Pet Owner



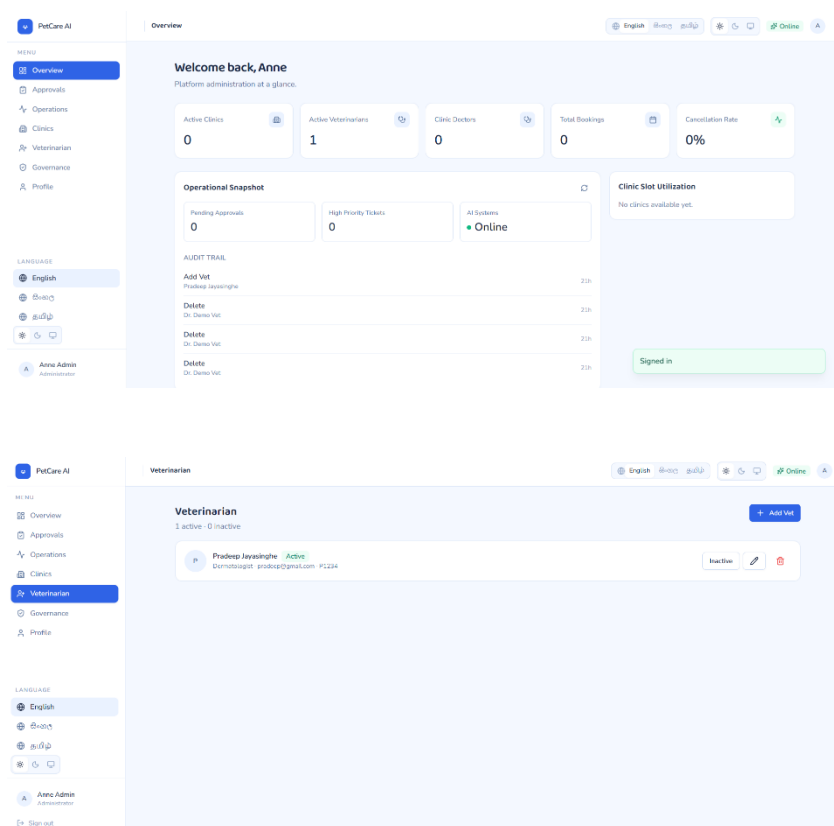


WEB – Veteranarian

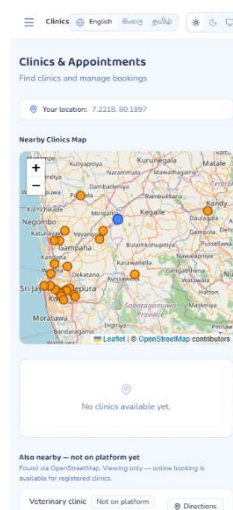
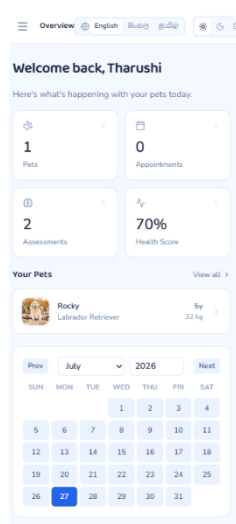
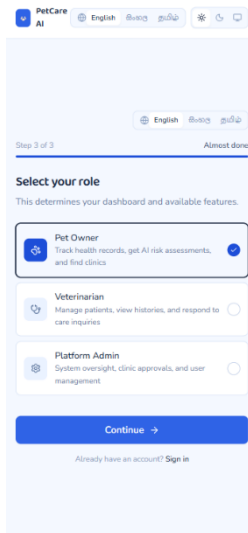




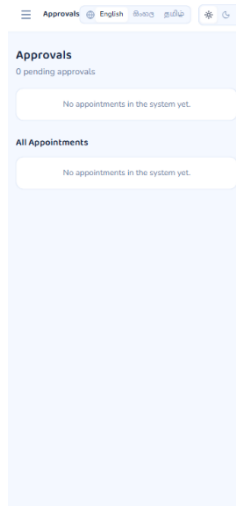
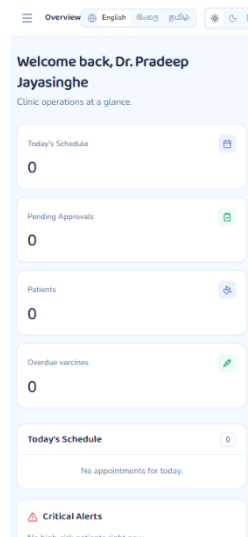
WEB – Admin



Mobile – Pet Owner



Mobile – Vet



Mobile – Admin

